

CHAPTER 7

REACTIONS

IN AQUEOUS

SOLUTIONS

By the end of this chapter you will have covered the following material.

Science Understanding

- The concentration of a solution is defined as the amount of solute divided by the amount of solution; this can be represented in a variety of ways including by the number of moles of the solute per litre of solution (mol L^{-1}) and the mass of the solute per litre of solution (g L^{-1}) ([ACSCH063](#))
- The presence of specific ions in solutions can be identified using analytical techniques based on chemical reactions, including precipitation and acid-base reactions ([ACSCH064](#))
- The pH scale is used to compare the levels of acidity or alkalinity of aqueous solutions; the pH is dependent on the concentration of hydrogen ions in the solution ([ACSCH066](#))
- Patterns of the reactions of acids and bases (for example, reactions of acids with bases, metals and carbonates) allow products to be predicted from known reactants ([ACSCH067](#))





Dreamstime/Walter Arce

Figure C7.1 ▲
Water is the most common solvent for natural solutions.

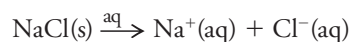
7.1 Aqueous solutions of ionic compounds

As you learnt in Chapter 2, solutions are defined as homogeneous mixtures. The most commonly occurring natural solutions have water as the solvent and a wide variety of different solutes, including solids, liquids and gases. Oceans are solutions of different compounds, many of which are ionic, dissolved in water. The chemical reactions that are responsible for maintaining all living things occur in water solutions. Solutions of substances dissolved in water are called **aqueous solutions**. The term 'aqueous' comes from the Latin *aqua*, meaning water.

In Chapter 6, you looked at how the nature of the solute and solvent and temperature affects the solubility of different substances. You also looked at the processes that occur when a substance dissolves, and different measures for the amount of substance dissolved.

When ionic substances dissolve in water, they dissociate. This means they separate into their ions, which are then able to move freely and independently of each other through the solution.

For example, when sodium chloride (NaCl) dissolves in water, it separates into Na^+ ions and Cl^- ions. This can be represented as:



Although most ionic compounds are soluble in water, they do not all dissolve to the same extent. The **solubility** of a substance is the degree to which it dissolves and is temperature dependent.

The solubility of a solute is the maximum amount in grams that can dissolve in 100 g of the solvent at a given temperature. The standard temperature for measuring solubility is 25°C.

Chemists use the terms 'soluble', 'insoluble' and 'sparingly soluble' to categorise the solubility of different compounds. 'Soluble' means that a compound dissolves to more than 10 g L⁻¹ (or 1 g/100 mL), 'insoluble' means that it dissolves to less than 1 g L⁻¹ and 'sparingly soluble' means that it dissolves between the range 1 g L⁻¹ to 10 g L⁻¹.

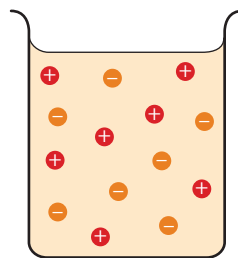


Figure C7.2 ►
A solution of an ionic substance

Precipitation reactions

When solutions of certain ionic compounds are mixed, they sometimes react to produce a solid called a **precipitate**. This type of reaction is called a **precipitation reaction**.

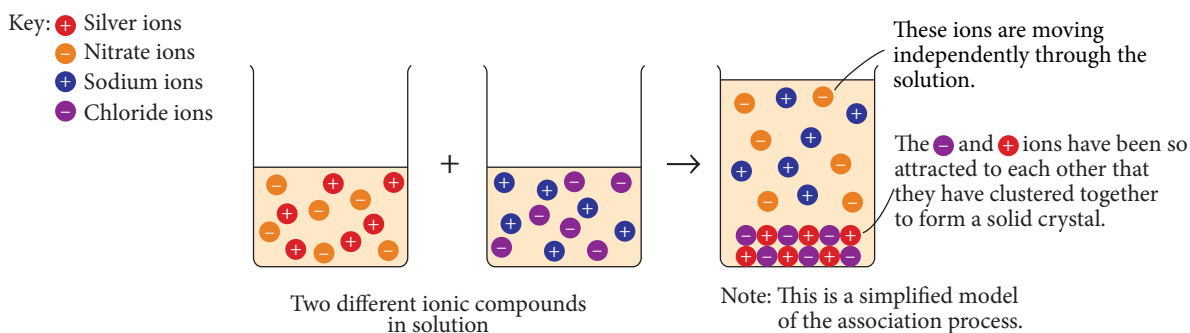
A precipitate is a solid produced by the reaction between two solutions.

For example, when sodium chloride solution is added to silver nitrate solution, a white solid of silver chloride is formed. When the solution of Na^+ and Cl^- ions is mixed with the solution of Ag^+ and NO_3^- ions, the result is a solution containing all the ions moving randomly throughout (i.e. no precipitate is formed). Remember that each of the ions is surrounded by water molecules. The Ag^+ and Cl^- ions are more strongly attracted to each other than they are to the water, so they cluster together in groups and form a solid, which precipitates out of solution, as shown in Figures C7.3 and C7.4.



Science Source/Charles D. Winters

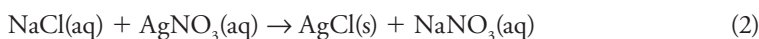
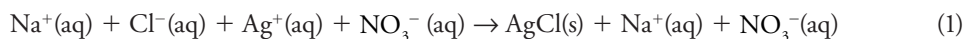
► **Figure C7.3**
A silver chloride precipitate



▲ **Figure C7.4**
Explaining the formation of silver chloride precipitate

The Na^+ and NO_3^- ions are not involved in the reaction and remain in the solution as ions. Ions that are not involved in a reaction are called **spectator ions**.

The equation for this reaction can be written in a number of different ways, which may include or omit the spectator ions.



Equation 1 is a **complete ionic equation**. It shows all the ions present in the solution. Equation 2 is an overall equation. It shows all the reactants and products as actual neutral compounds. Equation 3 is the **net ionic equation**. It shows only the reacting ions. Removing the spectator ions from the complete ionic equation produces the net ionic equation. This equation most accurately represents the reaction taking place, so is most commonly used to represent precipitation reactions.

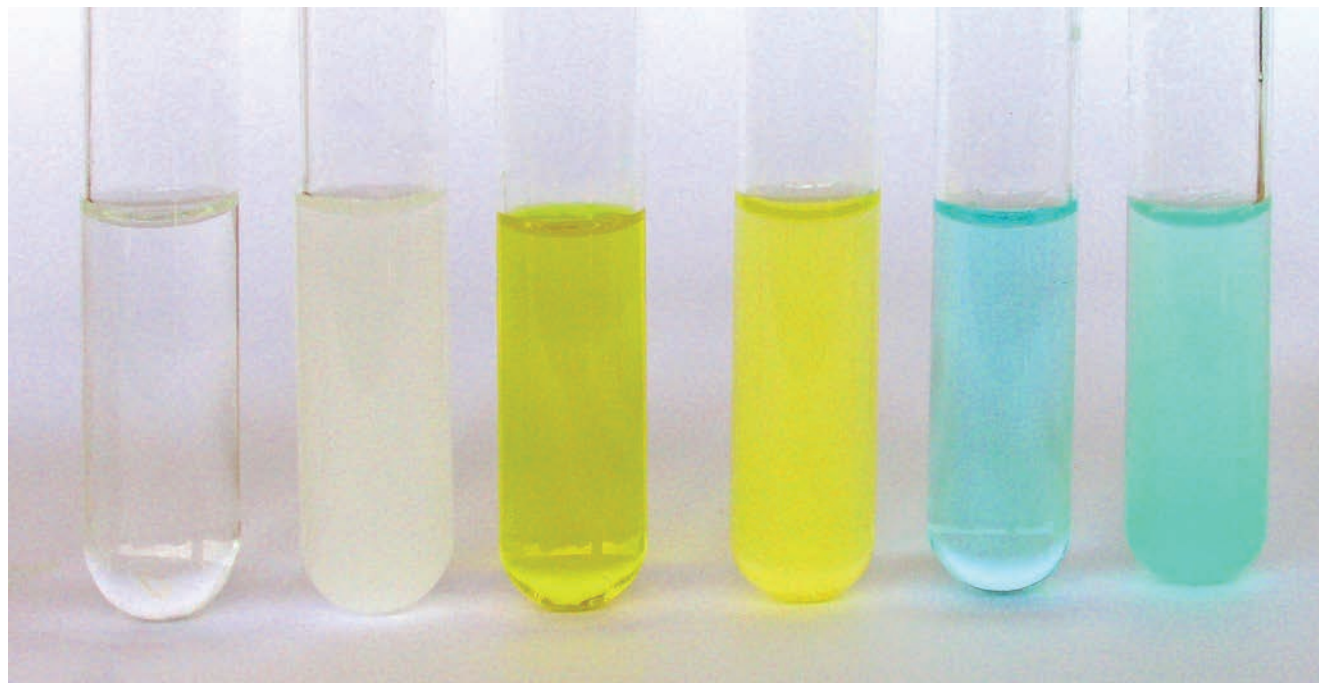


PRECIPITATION ANIMATION

View this animation and predict when a precipitate will form.

Predicting precipitation

By systematically mixing solutions that contain known cations and anions, chemists have experimentally determined which compounds are soluble and which form a precipitate (are insoluble). These results have been summarised into a table of solubility data that can then be used to predict whether or not precipitation will occur when two known solutions are mixed. Sometimes, when only a small amount of precipitate forms, it does not settle to the bottom but stays in suspension and makes the mixture look cloudy.



Roland Smith

Figure C7.5 ▲
Solutions and suspensions

EXPERIMENT 7.1

INVESTIGATING PRECIPITATION REACTIONS

Solubility rules can be formulated from experimental results. In this experiment, you will determine which mixtures of solutions form precipitates and use these results to construct solubility rules. These can then be compared with theoretical data.

Aim

To compare the solubility of a range of ionic substances through precipitation reactions

Materials

- Cations (all 0.1 mol L^{-1}) in dropper bottles: sodium nitrate (NaNO_3), silver nitrate (AgNO_3), lead(II) nitrate ($\text{Pb}(\text{NO}_3)_2$), ammonium nitrate (NH_4NO_3), magnesium sulfate (MgSO_4), iron(II) sulfate (FeSO_4), copper(II) sulfate (CuSO_4), zinc sulfate (ZnSO_4), aluminium sulfate ($\text{Al}_2(\text{SO}_4)_3$), barium chloride (BaCl_2), calcium chloride (CaCl_2), iron(III) chloride (FeCl_3)
- Anions (all 0.1 mol L^{-1}) in dropper bottles: potassium nitrate (KNO_3), potassium chloride (KCl), sodium sulfate (Na_2SO_4), sodium carbonate (Na_2CO_3), sodium acetate (NaCH_3COO), sodium hydroxide (NaOH)
- Distilled water
- 6 test tubes and test-tube rack
- Paper towel
- Test-tube cleaner

What are the risks in doing this experiment?	How can you manage these risks to stay safe?
Lead and barium salts are poisonous.	Wear safety glasses and avoid contact with skin. If contact occurs, wash thoroughly with soap and water.
Lead, barium, silver and copper salts are harmful to the environment.	Do not pour solutions containing these salts down the drain. Dispose of as directed by your teacher.

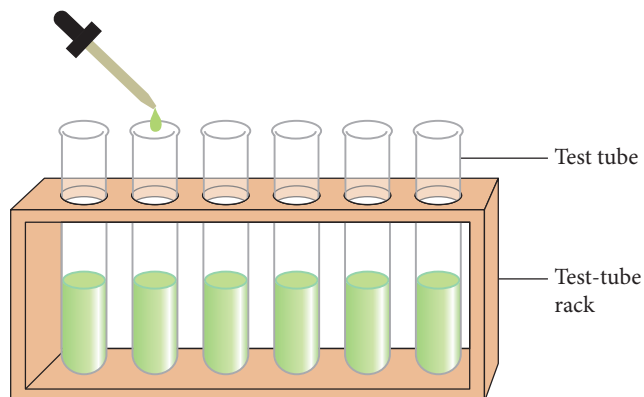
In your write-up, add any more risks you can think of, as well as ways to manage them. In particular, expand on the two risks listed to identify specific risks involved with each of them.

Procedure

- 1 Put a few drops of each anion into each of six test tubes. Take care not to mix droppers.
- 2 Add a few drops of the first cation to each. Record your observations.
- 3 Clean all test tubes thoroughly because incorrect results can arise from contamination of solutions.
- 4 Repeat step 2 for the next cation. Continue until all cations have been tested.

Results

Draw up a table similar to the one below to record your observations. For no reaction, record 'NR'; for a precipitate, record 'P' and the colour of the precipitate.



▲ Figure C7.6
Experimental set-up

Anion	Cation					
	NO_3^- nitrate	Cl^- chloride	SO_4^{2-} sulfate	CO_3^{2-} carbonate	CH_3COO^- ethanoate	OH^- hydroxide
Na^+						
Ag^+						
Pb^{2+}						
NH_4^+						
Mg^{2+}						
Fe^{2+}						
Cu^{2+}						
Zn^{2+}						
Al^{3+}						
Ba^{2+}						
Ca^{2+}						
Fe^{3+}						

Analysing the results

Summarise your results into statements such as 'All nitrates are . . .'

Discussion

- 1 **a** Compare the results of the experiment with the solubility rules in Table C7.1.
b What, if any, were the discrepancies between your results and Table C7.1? Suggest a reason for these discrepancies and a way of resolving them.
- 2 Write a balanced net ionic equation for each of the reactions in which a precipitate was obtained.

Conclusion

Summarise the solubility rules according to your results.

Table C7.1 summarises the solubility of common ionic compounds. The table applies to the cations in groups 1 and 2, NH_4^+ , Al^{3+} , Cr^{3+} , Mn^{2+} , Fe^{2+} , Fe^{3+} , Cu^{2+} , Co^{2+} , Ni^{2+} , Ag^+ , Zn^{2+} , Sn^{2+} , Cd^{2+} and Pb^{2+} . The table is organised in terms of the solubility of various anions. An additional useful generalisation is that all group 1 and NH_4^+ salts are soluble.

Table C7.1 Solubility of common ionic substances

Soluble anions	Exceptions
NO_3^-	None
CH_3COO^-	Ag^+ slightly soluble
Cl^-	Ag^+ insoluble; Pb^{2+} slightly soluble
Br^-	Ag^+ insoluble; Pb^{2+} slightly soluble
I^-	Ag^+ , Pb^{2+} insoluble
SO_4^{2-}	Ba^{2+} , Pb^{2+} , Sr^{2+} insoluble; Ag^+ , Ca^{2+} slightly soluble
Insoluble anions	Exceptions
OH^-	Group 1, NH_4^+ , Ba^{2+} , Sr^{2+} soluble; Ca^{2+} slightly soluble
O^{2-}	Group 1, NH_4^+ , Ba^{2+} , Sr^{2+} , Ca^{2+} soluble
S^{2-}	Groups 1 and 2, NH_4^+ soluble
CO_3^{2-}	Group 1, NH_4^+ soluble
SO_3^{2-}	Group 1, NH_4^+ soluble
PO_4^{3-}	Group 1, NH_4^+ soluble

WOW

Cyanide (CN^-)

Cyanide (CN^-) has long been used as a poison. In many crime novels, cyanide is chosen as the poison and cyanide 'suicide pills' were kept on hand by spies in the event of their capture. Leading Nazi Hermann Goering resorted to these after being tried at Nuremberg and found guilty of war crimes after World War II.

Applying solubility data

Chemists are often asked to identify substances such as cations and anions that may have contaminated food, soil and water. Precipitation reactions are useful for finding out whether a particular anion or cation is present; for example, chemists could use precipitation reactions to test for lead or cadmium in water supplies. Precipitation reactions can also be used to remove unwanted substances from water.

The data in Table C7.1 can be used to predict whether or not a precipitate will form when particular solutions are mixed.

Figure C7.7 ►
Chemists working in an analytical laboratory



Thinkstock/Horsche

WORKED EXAMPLE 7.1

- a What precipitate, if any, will form when aqueous solutions of magnesium iodide and silver nitrate are mixed? If a precipitate forms, write the neutral species and net ionic equations for the reaction.
- b Select a reagent that could be used to precipitate the cation in BaBr_2 , explaining your selection.

Answers

- a Precipitate is AgI.

Neutral species equation:



Net ionic equation:



- b Na_2CO_3 can be used to precipitate Ba^{2+} from a solution of BaBr_2 .

Logic

- 1 Write the formula of the compounds present.

Magnesium iodide (MgI_2)

Silver nitrate (AgNO_3)

- 2 Predict the products by swapping the partners of the reacting compounds and check that the formulas are correct.



- 3 Use Table C7.1 to see if either of the products is insoluble.

All nitrates are soluble, all iodides are soluble except Ag^+ and Pb^{2+} so AgI will precipitate

- 4 Write the required equations, checking they are balanced.

Neutral species equation is:



Net ionic equation is:



- 1 Identify the ions present and one to be precipitated.

Ba^{2+} and Br^- are the ions present in solution. Ba^{2+} is the ion to be precipitated.

- 2 Use Table C7.1 to identify an anion which could be used to precipitate the identified cation.

From Table C7.1, anions that will produce an insoluble compound with Ba^{2+} are S^{2-} , CO_3^{2-} , SO_3^{2-} and PO_4^{3-} .

- 3 Choose one of the anions and identify a cation to form a compound. Check that the cation does not react with the anion from the original compound.

Generally it is best to choose a cation from group 1 or NH_4^+ because compounds of these are soluble. It would be incorrect to choose Ag^+ and Pb^{2+} as they precipitate with Br^- .

- 4 Identify the reagent and write the formula.

Na_2CO_3 can be used to precipitate Ba^{2+} from a solution of BaBr_2 .

Try these yourself

- a When solutions of potassium sulfide and cobalt acetate are mixed, which salt, if any, will precipitate? If a precipitate does form, write the neutral species and net ionic equations.
- b Name and give the formula of the precipitate (if any) that forms when each of the following pairs of substances is mixed.
- Copper(II) chloride and potassium hydroxide
 - Sodium carbonate and ammonium sulfate
 - Lead nitrate and copper(II) sulfate
- c What solution would you mix with calcium chloride to obtain a precipitate of the cation?

INVESTIGATION 7.1

IDENTIFYING IONIC COMPOUNDS

The labels on the bottles of five different chemicals have fallen off. The bottles have been labelled A, B, C, D and E. The chemicals are known to be ammonium hydroxide, potassium iodide, sodium hydroxide, silver nitrate and zinc sulfate. You have been asked to identify what is in each bottle.

What is your aim?

Think about what you want to find out in this investigation.

Hint: Look at the all the possible pairings of the chemicals listed and use solubility rules to predict what precipitates will actually form.

What will you need?

Think about the equipment you will need. Will you need any additional chemicals to confirm your results?

Be specific about quantities of chemicals and don't forget to include basic materials like distilled water and paper towel.

What are the risks?

Construct a table similar to the one below. Identify specific risks involved in the investigation and ways that you will manage the risks to avoid injuries or damage to equipment. Ask your teacher to check your risk assessment before you proceed.

What are the risks in doing this investigation?	How can you manage these risks to stay safe?

How will you carry out the investigation?

Hint: You need to use solubility rules. Write your method in a logical sequence of steps. Don't forget to be specific and include quantities.

What results will you collect?

Refer back to your aim. Refer to the results of Experiment 7.1 for possible table structure and use those results to help identify products.

What have you found?

Consider what your results show. Look at theoretical predictions you made about possible precipitates and compare your results with these. Are there any differences or inconclusive results? Do you need to repeat any tests? Do you need to use an alternative solution for confirmation?

What do you conclude?

Reflect on what you found. Could you identify each of the solutions?

How could you improve the effectiveness of the identification process?

QUESTION SET 7.1

Remembering

- 1 Define 'precipitate'.
- 2 Explain what is meant by 'spectator ion'.

Understanding

- 3 Which of the following substances are soluble in water?
 - a Sodium hydroxide
 - b Aluminium oxide
 - c Copper(II) sulfate
 - d Silver carbonate
 - e Lithium chloride
- 4 Name a different pair of substances (using suitable amounts) that you could dissolve in the one sample of water to produce a solution identical with that containing:
 - a potassium chloride and zinc nitrate.
 - b ammonium sulfate and sodium carbonate.
 - c magnesium bromide and copper(II) sulfate.

Applying

- 5 Write overall equations for the reactions (if any) that occur when solutions of the following pairs of substances are mixed. If there is no reaction, write 'NR'.
 - a Sodium chloride and silver nitrate
 - b Copper(II) sulfate and potassium hydroxide
 - c Nickel chloride and potassium sulfate
 - d Sodium carbonate and iron(II) sulfate
 - e Zinc nitrate and ammonium sulfide
 - f Potassium carbonate and calcium chloride
- 6 Solutions of what substances would you mix to prepare the following compounds by precipitation? Write equations for your reactions.
 - a Magnesium carbonate
 - b Lead sulfate
 - c Silver bromide

Analysing

- 7 Separate portions of an unknown solution A formed precipitates when potassium carbonate and sodium hydroxide solutions were added to them, but did not give precipitates when solutions of sodium chloride or ammonium sulfate were added. Solution A could have contained:
 - A Pb^{2+} , Zn^{2+} , Cu^{2+}
 - B Mg^{2+} , Al^{3+} , Zn^{2+}
 - C Mg^{2+} , Ba^{2+} , Al^{3+}
 - D Cu^{2+} , Zn^{2+} , Ag^{+}
- 8 Describe tests to determine whether a solution contained:
 - a lead nitrate or barium nitrate.
 - b copper(II) sulfate or iron(II) sulfate.

7.2 Determining concentration

Aqueous solutions can contain many different dissolved substances. The amount of dissolved solute is regularly monitored in many areas of chemistry, including wine production, water quality and swimming pools. In many of these areas, it is important to know the **concentration** of the solute in the solution.

The concentration of a solution is the amount of solute present in a specified amount of solution.

Figure C7.8 ►

Monitoring water quality involves a range of measurements.



istockphoto/BarrCo

MOLARITY

Use this simulation to learn more about the relationship between moles of substance and volume of solution.



CONCENTRATION

Use this simulation to learn more about factors that cause concentration to change.



In Chapter 6, you learnt about some different measures of concentration (% w/w, % v/w, % w/v and ppm), which could be used to compare quantities of different substances in solution. Different ways of expressing concentration are used because each has its advantages in different situations.

Molarity

When calculating quantities in chemical reactions, the basic unit is the mole. Therefore, it is convenient to have a measure of concentration in terms of moles. This measure of concentration is called **molarity**. The molarity of a solution is defined as the number of moles of solute per litre of solution.

The concentration of a particular solution is given by:

$$\text{Concentration (mol L}^{-1}\text{)} = \frac{\text{number of moles of solute (mol)}}{\text{volume of solution (L)}}$$
$$c = \frac{n}{V}$$

This formula can also be rearranged to find n and V .

This is also referred to as molar concentration and given the symbol M . In your chemistry experiments, such as Experiment 7.1, you may use a solution that is labelled as 0.1 M sodium chloride, for example. This means 0.1 moles of sodium chloride per litre of solution.

Sometimes, chemists may need to know the mass of solute in a solution so units of g L^{-1} may also be used. In Chapter 5, you learnt the relationship between moles and mass:

$$n = \frac{m}{M}$$

Worked example 7.2 shows how moles, concentration and mass are calculated.

Be careful not to confuse M for molar mass in grams with M for molarity, which is used for solutions.

WORKED EXAMPLE 7.2

- a Calculate the concentration of copper(II) sulfate in a solution if 250 mL of the solution contains 0.01 mol of copper(II) sulfate.
- b What is this concentration in g L⁻¹?
- c Calculate the concentration of a solution prepared by dissolving 13.5 g of sodium hydroxide in 100 mL of solution.
- d What mass of silver nitrate is needed to prepare 250 mL of a 0.120 mol L⁻¹ solution of silver nitrate?

Answers

- a Concentration is 0.040 mol L⁻¹.

Logic

- 1 Write the relationship.

$$c = \frac{n}{V}$$

- 2 Identify the quantities and check units.

$$c = ?, n = 0.01 \text{ mol}, V = 250 \text{ mL} = 0.25 \text{ L}$$

- 3 Substitute to find c .

$$c = \frac{0.01}{0.25} = 0.040 \text{ mol L}^{-1}$$

- b Concentration is 6.4 g L⁻¹.

Convert moles to grams using $n = \frac{m}{M}$.

$$n = 0.040 \text{ mol}, m = ?, M(\text{CuSO}_4) = 63.55 + 32.06 + 4 \times 16.0 = 159.6 \text{ g mol}^{-1}$$

$$0.040 = \frac{m}{159.6}$$

$$m = 0.040 \times 159.6 = 6.4 \text{ g}$$

Concentration is 6.4 g L⁻¹

- c Concentration is 3.37 mol L⁻¹.

- 1 Write the relationships.

$$c = \frac{n}{V}, n = \frac{m}{M}$$

- 2 Identify the quantities and check units.

$$c = ? \text{ mol L}^{-1}, n = ?, V = 100 \text{ mL} = 0.100 \text{ L}, m = 13.5 \text{ g}$$

- 3 Calculate M and substitute to find n .

$$M(\text{NaOH}) = 40 \text{ g mol}^{-1}$$

$$n = \frac{13.5}{40} = 0.3375 \text{ mol}$$

- 4 Substitute to find c .

$$c = \frac{0.3375}{0.100} \text{ mol L}^{-1}$$

$$c = 3.38 \text{ mol L}^{-1}$$

- d Mass is 5.09 g.

- 1 Write the relationships.

$$c = \frac{n}{V}, n = \frac{m}{M}$$

- 2 Identify the quantities and check units.

$$c = 0.120 \text{ mol L}^{-1}, n = ?, V = 250 \text{ mL} = 0.25 \text{ L}, m = ?$$

- 3 Substitute to find n .

$$0.120 = \frac{n}{0.25} \text{ mol L}^{-1}$$

$$n = 0.120 \times 0.25 = 0.03 \text{ mol}$$



4 Calculate M and substitute to find m .

$$M(\text{AgNO}_3) = 169.9 \text{ g mol}^{-1}$$

$$0.03 = \frac{m}{169.0}$$

$$m = 0.03 \times 169.9 = 5.09 \text{ g}$$

Try these yourself

- a Calculate the molarity of:
- 0.20 mol of zinc iodide dissolved in 1.5 L of solution.
 - 0.04 mol of magnesium chloride dissolved 450 mL of solution.
- b Calculate the concentration of the following solutions made by dissolving a known mass in a given volume.
- 8 g potassium chloride dissolved in 250 mL of solution
 - 1.46 g iron(II) sulfate dissolved in 100 mL of solution
 - 7.5 g silver nitrate dissolved in 500 mL of solution
- c What mass of solute is needed to make the following solutions?
- 2.00 L of a 0.200 mol L^{-1} solution of sodium iodide
 - 250 mL of a 0.01 mol L^{-1} solution of potassium phosphate
 - 500 mL of a $0.0075 \text{ mol L}^{-1}$ solution of barium nitrate

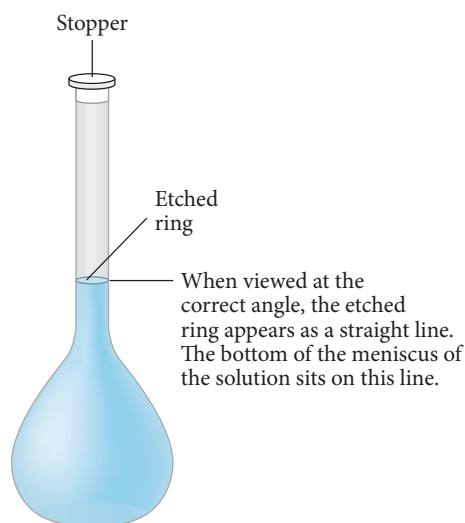


Figure C7.9 ▲
A solution prepared in a volumetric flask

Laboratory preparation of a solution

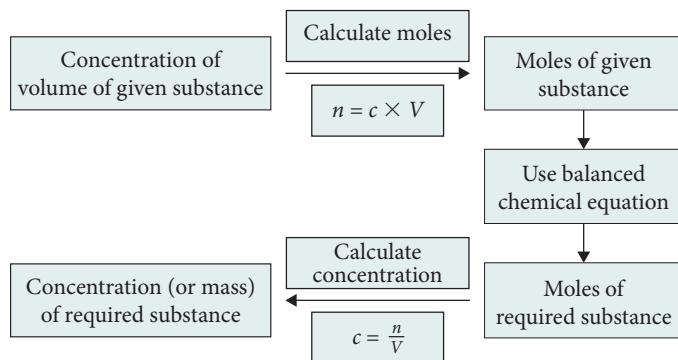
To prepare a solution of known concentration, chemists must use precise apparatus. The volume of the final solution must be as accurate as possible, so a **volumetric flask** is used because these are designed to allow precise volume determinations.

The steps for preparing a solution of known concentration are as follows.

- 1 The required amount of solute is weighed and then transferred to a volumetric flask of required volume.
- 2 The beaker that contained the solute is rinsed with distilled water and the rinse water is transferred to the volumetric flask.
- 3 More water is added to dissolve the solute and then up to the calibration mark.
- 4 The stopper is placed in the flask, which is inverted several times to ensure thorough mixing of the solution.

Calculating reacting quantities

Now that a relationship between moles and concentration has been developed, we can extend calculations involving chemical reactions to include those involving solutions.



◀ **Figure C7.10**
Calculating reaction quantities

WORKED EXAMPLE 7.3

What mass of barium sulfate will be formed when excess sodium sulfate is added to 50 mL of a 0.30 mol L⁻¹ solution of barium chloride?

Answer

Mass is 3.5 g.

Logic

1 Write a balanced equation.



2 Calculate the number of moles of BaCl₂.

$$c = \frac{n}{V}$$

$$c = 0.30 \text{ mol L}^{-1}, V = 50 \text{ mL} = 0.050 \text{ L}$$

$$0.30 = \frac{n}{0.050}$$

$$n = 0.30 \times 0.050 = 0.015 \text{ mol}$$

3 Use the balanced equation to calculate the number of moles of BaSO₄.

$$n(\text{BaSO}_4) = n(\text{BaCl}_2) = 0.015 \text{ mol}$$

4 Calculate the mass of BaSO₄.

$$n = \frac{m}{M}$$

$$n = 0.015 \text{ mol}, m = ?, M(\text{BaSO}_4) = 233 \text{ g mol}^{-1}$$

$$0.015 = \frac{m}{233}$$

$$m = 0.015 \times 233 = 3.5 \text{ g}$$

Try these yourself

- What mass of copper(II) hydroxide is formed when excess copper(II) sulfate is added to 100 mL of a 0.450 mol L⁻¹ solution of sodium hydroxide?
- Calculate the mass of magnesium carbonate formed when excess sodium carbonate is added to 75 mL of a 0.08 mol L⁻¹ solution of magnesium chloride.
- If excess NaOH is added to 10 mL of a 0.500 mol L⁻¹ solution of zinc chloride, calculate the mass of Zn(OH)₂ produced.



WOW

CONCENTRATION CALCULATIONS

To practise more problems involving concentration calculations, complete this molarity worksheet.

Hard or soft water

The presence of magnesium and calcium ions affects the taste and usefulness of water. These ions occur naturally in water and the concentration varies with the source and contributes to the hardness of water. Water is described as 'hard' if it is difficult to form a lather or suds when washing. This is because the calcium and magnesium ions react with soaps and detergents to form an insoluble scum, making them ineffective for cleaning.

QUESTION SET 7.2

Remembering

- 1 What is the relationship between molarity and concentration?

Understanding

- 2 Calculate the concentration of the following solutions in
- i mol L^{-1}
 - ii g L^{-1}
- a 2.00 L of solution containing 1.50 mol of potassium bromide
 - b 250 mL of solution containing 0.0025 mol of sodium iodide
- 3 Calculate the number of moles of solute that is needed to make the volume of the solution in the following.
- a 100 mL of a 0.40 mol L^{-1} solution of barium nitrate
 - b 500 mL of a 0.15 mol L^{-1} solution of sodium sulfate
 - c 2.00 L of a 0.032 mol L^{-1} solution of copper(II) chloride

Applying

- 4 What mass of solute is required for each of Questions 3a–c?
- 5 Calculate the molar concentration of the following solutions.
- a 199 g NiBr_2 dissolved in 5.00 L of solution
 - b 0.059 g KF dissolved in 227 mL of solution
 - c 23 g ethanol ($\text{C}_2\text{H}_5\text{OH}$) in 750 mL of solution
- 6 What mass of solute is needed to make the following solutions?
- a 25.0 mL of a 0.02 mol L^{-1} solution of sodium carbonate
 - b 32.0 mL of a 0.10 mol L^{-1} solution of copper(II) sulfate
- 7 Calculate the volume of a 0.025 mol L^{-1} solution that contains $1.0 \times 10^{-3} \text{ mol}$ of magnesium nitrate.

Analysing

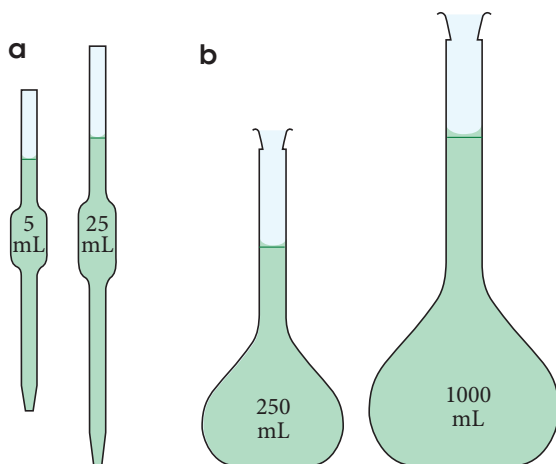
- 8 What mass of silver iodide will be formed when excess sodium iodide is added to 25 mL of a 0.15 mol L^{-1} solution of silver nitrate?
- 9 To measure the concentration of calcium ions in a water sample, an analyst took 500 mL of the sample and added sodium carbonate until no further precipitation occurred. The precipitate was filtered off, dried and weighed. It was found to have a mass of 1.72 g.
- a What was the molarity of calcium ions in the original solution?
 - b How many grams of calcium ions were there per litre in the original solution?
 - c Water is considered to be 'hard' if it contains more than $201\text{--}300 \text{ mg L}^{-1}$ calcium carbonate. Would this water sample be classified as 'hard'?

7.3 Dilutions

In order to reduce transport and packaging costs, many products such as cordials, detergents, laboratory chemicals and pesticides are sold in a concentrated form. Water is added when they are ready for use. The amount of water added to cordials depends on personal preference, but for pesticides there are instructions on the packaging for making solutions of specific concentrations. This is to ensure the final mixture is suitable for its intended purpose. The process of adding water to a solution to make it less concentrated is called **dilution**.

Dilution is an important laboratory technique used to produce solutions of required concentration for testing and analysis procedures. The dilution process must be as accurate as possible, so the concentration of the required diluted solution is correct.

The usual equipment to ensure accurate measures of volume are pipettes and volumetric flasks similar to those shown in Figure C7.11.



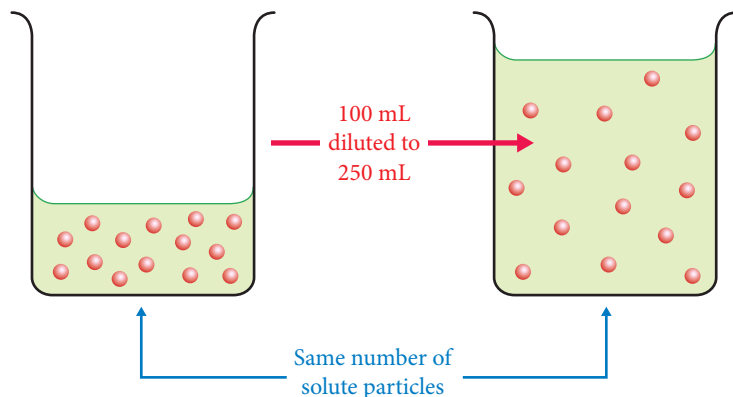
◀ **Figure C7.11**

a) Pipettes and b) volumetric flasks are used in the laboratory when accurate volumes are required.

The dilution procedure involves:

- 1 using a pipette to measure a definite volume of a solution of known concentration
- 2 transferring this volume to a volumetric flask
- 3 adding sufficient water to the volumetric flask to make up the solution to the calibration mark.

To calculate the concentration of the diluted solution, we use the fact that the number of moles (particles) in the diluted solution is the same as in the original volume of solution.



Never add water to a concentrated acid solution because the heat generated by the reaction could cause the concentrated acid to boil and spit out of the container. It can also cause the glass container to break. The correct way to dilute a concentrated acid is to add small quantities of the acid to large quantities of water so the heat generated is absorbed by the water.

◀ **Figure C7.12**

Dilution: the same number of moles in a larger volume of solvent



DILUTIONS

To practise more problems involving dilutions, complete this dilutions worksheet.

Because the number of moles of solute remains constant, we can use the following formula to relate concentration and volume of the original solution to that of the diluted solution. This relationship is called the dilution formula:

$$c_1V_1 = c_2V_2$$

where c_1 and c_2 are the initial and final concentrations of the solution in the same units, and V_1 and V_2 are the initial and final volumes of the solution, in the same units.

WORKED EXAMPLE 7.4

- a** What volume of 12 mol L^{-1} hydrochloric acid is required to prepare 500 mL of a 0.25 mol L^{-1} solution?
- b** What is the concentration of a solution made by diluting 20 mL of a 5.0 mol L^{-1} solution to 1.00 L ?

Answers

- a** The volume is 10.4 mL .

Logic

- 1** Write the relationship.

$$c_1V_1 = c_2V_2$$

- 2** Identify the quantities and check units for consistency.

$$c_1 = 12 \text{ mol L}^{-1}, V_1 = ?, c_2 = 0.25 \text{ mol L}^{-1}, V_2 = 500 \text{ mL}$$

- 3** Substitute.

$$12 \times V_1 = 0.25 \times 500$$

$$V_1 = \frac{0.25 \times 500}{12} = 10.4 \text{ mL}$$

- b** The concentration is 0.10 mol L^{-1} .

- 1** Write the relationship.

$$c_1V_1 = c_2V_2$$

- 2** Identify the quantities and check units for consistency.

$$c_1 = 5.0 \text{ mol L}^{-1}, V_1 = 20 \text{ mL}, c_2 = ?, V_2 = 1.00 \text{ L} = 1000 \text{ mL}$$

- 3** Substitute.

$$5.0 \times 20 = ? \times 1000$$

$$c_2 = \frac{5.0 \times 20}{1000} = 0.10 \text{ mol L}^{-1}$$

Try these yourself

- a** What volume of solution is needed to dilute 18 mol L^{-1} sulfuric acid to prepare 2.5 L of 0.1 mol L^{-1} sulfuric acid?
- b i** A student was preparing for an experiment and diluted 10 mL of 5 mol L^{-1} copper(II) nitrate solution to 200 mL . What is the concentration of the final solution?
- ii** After performing the dilution the student realised they had made a mistake because they actually required 100 mL of a 0.10 mol L^{-1} solution. How much of the solution they diluted in part **i** should they use to obtain the required solution?
- c** A concentrated solution (10.0 mL) of sodium hydroxide was mixed with 55 mL of water to produce a solution of concentration 0.48 mol L^{-1} . What was the concentration of the original solution?

EXPERIMENT 7.2

PREPARING AND DILUTING SOLUTIONS

Colorimetry is an analytical technique used to determine the concentration of an unknown solution by comparing its colour with those of solutions of known concentration. In this experiment, you will prepare solutions of known concentration and use these to determine the concentration of a solution of unknown concentration.

Aim

To prepare solutions of known concentration and use these to determine the unknown concentration of a solution

Materials

- Approx. 0.1 g potassium permanganate (KMnO_4)
- Solution of unknown concentration
- 250 mL beaker
- 25 mL pipette
- 4 × 250 mL volumetric flasks
- Stirring rod
- Filter funnel
- Electronic balance
- Water
- Paper towel

What are the risks in doing this experiment?	How can you manage these risks to stay safe?
Potassium permanganate can stain skin and clothing.	Wear protective clothing and avoid contact.
Potassium permanganate can be irritating to the skin and eyes.	Wear safety glasses and avoid contact with the chemical.

In your write-up, add any more risks you can think of, as well as ways to manage them. In particular, expand on the two risks listed to identify specific risks involved with each of them.

Procedure

- 1 Weigh out approximately 0.1 g of potassium permanganate in one of the beakers. (Make sure you know the mass you are using accurately.)
- 2 Add just enough water to dissolve the crystals of potassium permanganate and transfer the solution to a 250 mL volumetric flask.
- 3 Fill the flask to the 250 mL mark with water and mix thoroughly.
- 4 Draw up a table like the one in the results section and complete it for this first solution.
- 5 Pipette 25 mL of the first solution into a clean volumetric flask and fill to 250 mL.
- 6 Complete the table for this second solution.
- 7 Repeat steps 5 and 6 using the second and third solutions so you have successive dilutions.
- 8 Compare the colour of your solution of unknown concentration to the colours of your diluted solutions.

Results

Solution	Relative colour intensity	Initial volume	Final volume	Mass KMnO_4	Moles KMnO_4	Concentration (mol L^{-1})
1						
2						
3						
4						

Analysing the results

- 1 Complete the results table.
- 2 What effect does dilution have on:
 - a colour intensity?
 - b concentration of the solution?
 - c mass of the solute in the diluted solution compared to the original mass?

Discussion

- 1
 - a What is the concentration of the unknown solution?
 - b How easy was it to determine this?
 - c What could be done to more accurately determine the unknown concentration?
- 2 How would you prepare a solution twice as concentrated as solution 2?

7.4 Acids and bases

Our environment is naturally acidic. Rain is slightly acidic as are many foods and drinks. Soft drink contains carbonic acid, citrus fruits contain citric acid and many foods contain vitamin C, which is ascorbic acid. Acids are also important components of many of the compounds that make up living things. Proteins are long chains of amino acids and our stomach produces hydrochloric acid to aid with digestion of food.

Bases are also part of our natural environment, although they are not as commonly recognised as acids. Ammonia is a base found in volcanic gases and produced by rotting plant and animal matter. Caffeine, found in coffee and tea, is a base as is nicotine, which is in tobacco.



Figure C7.13 ▲
Common substances containing a) acids and b) bases

Definitions of acids and bases

In your Years 9 and 10 science course, you probably studied acids and bases and learnt practical definitions of these substances. Practical definitions are ones based on the common properties of a substance. Acids and bases are families of substances so members of these families can be recognised by their common properties.

Common properties of acids include:

- a sour taste
- sting or burn the skin
- conduct electricity in solution
- turn blue litmus red.

Common properties of bases include:

- a slippery feel in aqueous solution
- a bitter taste
- conduct electricity in solution (not all bases are soluble)
- turn red litmus blue.

WOW

Acid rain

Acid rain is rain that has become acidic because gases from the atmosphere have dissolved in it. About 70% of acid rain comes from sulfur dioxide (SO_2), which dissolves into the water to form sulfuric acid. The rest comes from various oxides of nitrogen (mainly NO_2 and NO_3 , collectively called NO_x).

While the effect of acid rain on forests, lakes and rivers has been well publicised, the effects on people have not been as well noted. Many toxic metals are held in the ground in compounds. However, acid rain can break down some of these compounds, freeing the metals and washing them into water sources such as rivers. In Sweden, nearly 10 000 lakes now have such high mercury concentrations that people are advised not to eat fish caught in them.

The effects of acids and bases on vegetable dyes such as litmus was initially used to classify substances as acids and bases.

While practical definitions are useful in the laboratory, chemists attempt to define substances by what they are as well as by what they do. The definition of an acid has changed several times over the last 250 years. French scientist Antoine Lavoisier (1743–94) was one of the first to try to establish what an acid is. In the late 19th century, Swedish chemist Svante Arrhenius attempted to explain acids and bases in terms of the particles they produced in aqueous solution. He proposed that acids were substances that ionised in solution to produce hydrogen ions and that acids were strong if they ionised almost completely and weak if the ionised only slightly. Arrhenius defined a base as a substance that in solution produced hydroxide ions. However, not all bases dissolve in water so his definition has been modified to include insoluble bases.

The definition of a base is quite general. Not all oxides are basic. Most oxides formed between a non-metal and oxygen (for example CO_2 and SO_3) are acidic or neutral, while those formed between a metal and oxygen (for example CaO and MgO) are basic or **amphoteric**.

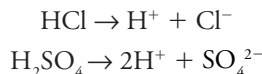
For present purposes, the following simple definition of acids and bases will be used.

You will learn about acids and bases in more detail in Units 3 & 4.

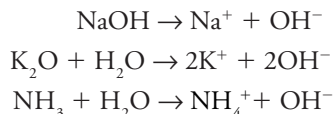
An **acid** is a substance that produces hydrogen ions (H^+) in aqueous solution. More strictly, the hydrogen ion attaches to a water molecule and forms **hydronium ions** (H_3O^+).

A **base** is a substance that either contains the oxide ion (O^{2-}) or hydroxide ion (OH^-) or produces hydroxide ion in aqueous solution. A soluble base is called an **alkali**.

Common acids are hydrochloric acid (HCl), sulfuric acid (H_2SO_4) and nitric acid (HNO_3). When these acids are mixed with water to form acidic solutions, the molecules dissociate:



Common soluble bases (alkalis) are sodium hydroxide ($NaOH$), barium hydroxide ($Ba(OH)_2$), potassium oxide (K_2O) and ammonia (NH_3). These substances either contain, or in aqueous solution produce, the hydroxide ion (OH^-). For example:



Acid–base reactions

Acids and bases react to form compounds called salts and water. The name given to this reaction is **neutralisation**.

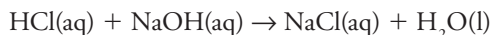
If the right amount of acid and base react, the resulting solution will be **neutral** (neither acidic nor basic).

The generalised reaction given above can be used to predict the products of most reactions between an acid and base.

A typical reaction between acids and bases is the reaction between a hydroxide and an acid:



For example:



It is usual to write these reactions using the overall equation. The Na^+ and Cl^- ions are spectator ions so the net ionic equation is: $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$

WOW

Neutralisation reactions

Neutralisation reactions are important in our bodies and in our lives. For example, if too much hydrochloric acid is produced in our stomachs, then we could end up with heart burn or indigestion. This problem is commonly fixed by taking an antacid, which contains a base such as magnesium hydroxide or aluminium hydroxide and neutralises the excess acid.

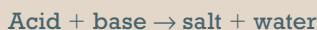
Figure C7.14 ►

Antacids contain a base, which neutralises excess acid in the stomach.



clibrary.com

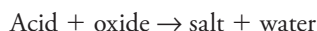
Neutralisation reactions can be written as the following generalised form:



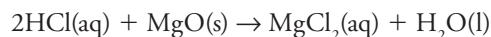
ANTACIDS

Visit this website to find out more about the chemistry of how antacids work.

Another typical reaction is that between a basic or amphoteric oxide and an acid:

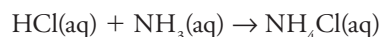


For example:

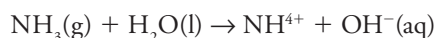


Another common but not typical acid–base reaction worth noting is the reaction between an acid and ammonia. This reaction does not adhere to the generalisation given above because the H^+ ion attaches to the NH_3 molecule (not H_2O) to produce $\text{NH}_4^+(\text{aq})$.

For example:



While ammonia is considered to be a weak base due the production of OH^- ions when it reacts with water,



the ammonium ion is considered to be a weak acid. This means it will react with a base to produce a salt and water. However, ammonia gas will also be produced in the reaction as seen in the following example:



Ammonium salts such as ammonium sulfate or ammonium nitrate are commonly used as fertilisers. This is one reason why a farmer wouldn't treat a field with an ammonium compound at the same time as using an alkali (a soluble base) such as lime. The ammonium salt will react with the base to give off ammonia, which will escape into the atmosphere as a gas, and so not be available to the plants.

QUESTION SET 7.3

Remembering

- 1 Define 'acid' and 'base'.
- 2 What is meant by neutralisation? Give an example.
- 3 What is the difference between the hydrogen and hydronium ions?

Understanding

- 4 Write the dissociation reaction for the following acids in water.
 - a Nitric acid (HNO_3)
 - b Hydrochloric acid (HCl)
 - c Phosphoric acid (H_3PO_4)
- 5
 - a Use Table C7.1 to determine which of the following oxides and hydroxides are soluble: $\text{Ca}(\text{OH})_2$, CuO , NH_4OH , Na_2O .
 - b Write the equation for the reaction of each with water.

Applying

- 6 Write balanced overall equations for the following reactions.
 - a Nitric acid with potassium hydroxide
 - b Sulfuric acid with zinc oxide
 - c Hydrofluoric acid with magnesium hydroxide
 - d Phosphoric acid with ammonia
 - e Acetic acid with sodium oxide
 - f Calcium oxide with ammonium nitrate



ACIDS AND BASES

Visit this online lab, which has many different resources showing properties and reactions of acids and bases as well as pH and indicators

Analysing

- 7 a Describe a method of preparing a solution of known concentration of the salt Na_2SO_4 .
b How could the salt be obtained from the solution?
c What needs to be taken into account in the preparation to ensure only this salt is obtained?

7.5 Indicators and pH

Litmus is a mixture of water-soluble dyes extracted from lichens. Acids and bases change the colour of litmus. Litmus turns red in acidic solutions, and blue in basic solutions. Litmus is called an acid–base **indicator**.

Indicators are commonly used to measure acidity and alkalinity in soil and water, such as swimming pools, as well as monitor waste from industries, laboratories and sewage treatment plants.

An indicator can also be used to provide information on the degree of acidity or alkalinity of a solution. However, because chemists need more precise measures of the degree of acidity and alkalinity of solutions, they have developed a scale to measure the concentration of acid in a solution. This scale is called the **pH scale**.



Figure C7.15 ▲

Checking the approximate pH of soil using an indicator colour chart

The pH scale

The term 'pH' stands for 'hydrogen power'. This name came about because the scale is based on the concentration of hydrogen ions in solution. Danish biochemist Soren Sorensen devised the scale to make writing the hydrogen ion concentration in solution more manageable.

The pH scale is based on the concentration of hydrogen ions in solution. It provides a measure of the acidity or alkalinity in aqueous solutions.

The pH scale generally ranges from 0 to 14. The lower the pH, the more acidic a solution is; the higher the pH, the more basic the solution is. Acids have a pH of less than 7. Bases have a pH greater than 7. A substance that has a pH equal to 7 is neutral.

Table C7.2 shows the relationship between pH and hydrogen ion concentration.

Table C7.2 pH and hydrogen ion concentration

pH	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
H⁺ concentration (mol L⁻¹)	1	10 ⁻¹ or 0.1	10 ⁻²	10 ⁻³	10 ⁻⁴	10 ⁻⁵	10 ⁻⁶	10 ⁻⁷	10 ⁻⁸	10 ⁻⁹	10 ⁻¹⁰	10 ⁻¹¹	10 ⁻¹²	10 ⁻¹³	10 ⁻¹⁴

Notice the relationship between the pH value and the superscript number in the hydrogen ion concentration. They are the same number because pH is based on the hydrogen ion concentration of a solution.

An important aspect of the pH scale is that a change in 1 unit on the scale is equivalent to a 10 times change in the hydrogen ion concentration. For example, a change from pH 1 to 2 is equivalent to a change in the hydrogen ion concentration from 0.1 to 0.01 mol L⁻¹.

Although the pH scale is based on the hydrogen ion concentration, it can also be used to determine the alkalinity and therefore the hydroxide ion (OH⁻) concentration. A pH of 7 represents a neutral solution and, as you learnt earlier, neutralisation occurs when an acid and base (alkali) react to produce a salt and water. Therefore, if equal concentrations of H⁺ and OH⁻ ions were to react, the resultant solution should be neutral. At pH 7, the concentration of H⁺ is 10⁻⁷ mol L⁻¹ so the concentration of OH⁻ must also be 10⁻⁷ mol L⁻¹.

Table C7.3 includes the concentration of OH⁻ ions compared to pH.

Table C7.3 pH and OH⁻ ion concentration

pH	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
H⁺ concentration (mol L⁻¹)	1 or 10 ⁰	10 ⁻¹ or 0.1	10 ⁻²	10 ⁻³	10 ⁻⁴	10 ⁻⁵	10 ⁻⁶	10 ⁻⁷	10 ⁻⁸	10 ⁻⁹	10 ⁻¹⁰	10 ⁻¹¹	10 ⁻¹²	10 ⁻¹³	10 ⁻¹⁴
OH⁻ concentration (mol L⁻¹)	10 ⁻¹⁴	10 ⁻¹³	10 ⁻¹²	10 ⁻¹¹	10 ⁻¹⁰	10 ⁻⁹	10 ⁻⁸	10 ⁻⁷	10 ⁻⁶	10 ⁻⁵	10 ⁻⁴	10 ⁻³	10 ⁻²	10 ⁻¹	10 ⁰

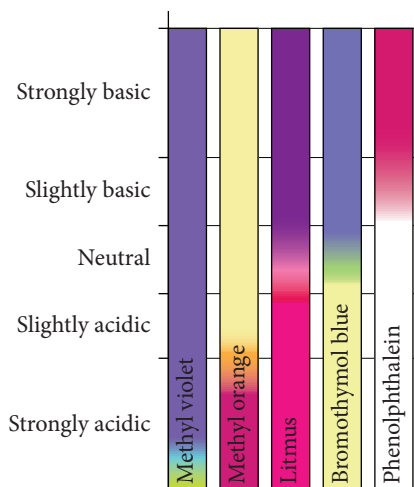
You should notice that as the H⁺ ion concentration decreases, the OH⁻ ion concentration increases. You should also notice that the concentration superscripts always add to 14.

In Units 3 & 4, you will learn more about this relationship and apply it to solve problems related to unknown concentrations of acids and bases.

Measuring pH

There are many natural and synthetic indicators that are used to indicate whether a solution is acidic or alkaline as well as the degree of acidity. Some synthetic indicators are methyl violet, methyl orange, litmus, bromothymol blue and phenolphthalein. Their colour changes are shown in Figure C7.16. The actual range of acidity or alkalinity over which an indicator changes colour varies from one indicator to another.

Chemists have developed a **universal indicator** that produces several colour changes depending on the pH of the solution. It is a mixture of several indicators.



Exponential notation is a better way of representing the numbers involved in the concentration of H⁺.

Neutralisation occurs when an acid and base (alkali) react to produce a salt and water. While this statement is correct, there are other factors related to the salt produced that will affect the pH of the final solution. You will learn more about this in Units 3 & 4.



pH SCALE AND CONCENTRATIONS

Visit this website to run a virtual lab that shows the relationship between the pH scale and acid and base concentrations.

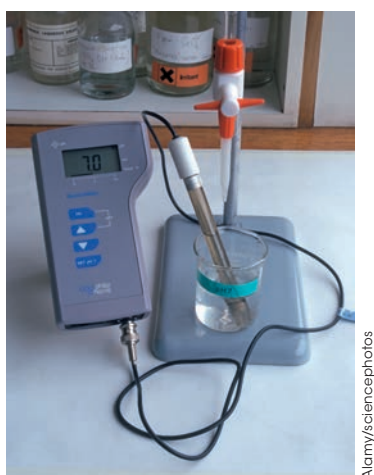
◀ **Figure C7.16** Colour changes in a range of indicators

Table C7.4 pH of some common substances

Colour of universal indicator	$[H_3O^+]$	pH	Substance	
	10	-1	Concentrated hydrochloric acid	ACID
	1	0	Car battery acid 1 mol L ⁻¹ hydrochloric acid	
	10 ⁻¹	1	0.1 mol L ⁻¹ hydrochloric acid	
	10 ⁻²	2	Stomach acid	
	10 ⁻³	3	Vinegar Lemon juice	
	10 ⁻⁴	4	Soft drinks Soda water	
	10 ⁻⁵	5	Wine Black coffee	
	10 ⁻⁶	6	Rain water Milk, saliva	
	10 ⁻⁷	7	Very pure water	NEUTRAL
	10 ⁻⁸	8	Blood Sea water	ALKALINE
	10 ⁻⁹	9	Bore water Baking soda solution	
	10 ⁻¹⁰	10	Toilet soap	
	10 ⁻¹¹	11	Laundry detergents	
	10 ⁻¹²	12	Household ammonia Dishwashing machine powders	
	10 ⁻¹³	13	Chlorine bleach solutions	
	10 ⁻¹⁴	14	Oven cleaners 1.0 mol L ⁻¹ sodium hydroxide	

Figure C7.17 ▼
A pH meter

Instruments such as pH meters and pH probes can be used to give a digital read-out of the pH of a solution.



Alamy/sciencephotos

EXPERIMENT 7.3

MEASURING pH

By measuring the pH of various substances, you can determine how acidic and basic they are. In this investigation, you will be able to determine the actual pH of these substances by using universal indicator and a pH meter. An advantage of using a pH meter is that it does not involve adding any substances to the solution and usually gives a more accurate pH provided the meter is calibrated correctly.

Aim

To measure and compare the pH of various substances

Materials

- Universal indicator solution and pH colour chart
- Distilled water
- Various household substances, such as vinegar, lemon juice, dish washer detergent, shampoo, antacid tablet, baking soda, vitamin C, bleach, washing detergent, fabric conditioner, soda water, salt, sugar
- Petri dishes (one for each soil sample)
- Soil samples including potting mix
- Barium sulfate (BaSO_4) powder
- Samples of water from swimming pools and local waterways
- Test tubes and test-tube rack
- Spatula
- 1 mol L^{-1} and $0.0001 \text{ mol L}^{-1}$ hydrochloric acid (HCl)
- 1 mol L^{-1} and $0.0001 \text{ mol L}^{-1}$ sodium hydroxide (NaOH)
- pH meter

What are the risks in doing this experiment?	How can you manage risks to stay safe?
Household chemicals can burn the skin.	Wear safety glasses and protective clothing. Take care when handling the chemicals and tell your teacher if there is a spill.
Food products can become contaminated in the laboratory.	Do not eat or drink any food used in the laboratory.
Soil samples may contain harmful micro-organisms.	Do not touch the soil sample. Work in a well-ventilated area. Wash your hands before eating.

In your write-up, add any more risks you can think of, as well as ways to manage them. In particular, expand on the two risks listed to identify specific risks involved with each of them.

Procedure

Part A: Testing household substances

- 1 Set up test tubes in the rack. Place a few millilitres (enough to measure with the pH meter) of the substance to be tested in a test tube. (You may need to add distilled water if the substance is a solid.)
- 2 Test the pH with the pH meter.
- 3 Rinse the pH meter with distilled water.
- 4 Add one drop of universal indicator solution.
- 5 Record the colour and pH.
- 6 Repeat with all other household substances.

Part B: Testing water samples

- 1 If the samples contain a lot of suspended sediment, filter or allow them to settle before testing.
- 2 Repeat the procedure used in part A to test the samples.

Part C: Testing soil samples

- 1 Place approximately half a spatula of soil in a Petri dish.
- 2 Add a few drops of indicator to make a paste and stir it.

- 3 Lightly sprinkle a thin layer of barium sulfate powder over the soil paste. Do not stir.
- 4 Allow a few minutes for the powder to absorb the indicator. Record the colour and pH values.
- 5 Repeat steps 1–4 with other soil samples.

Results

Draw up a data table to record the colour, pH from universal indicator and pH from the meter for each substance. Include another column for hydrogen ion concentration.

Analysis of results

- 1 Use Table C7.2 to determine the hydrogen ion concentration of each solution. You may need to record a range for this value if the reading was not a whole number.
- 2 Make a list of substances from the most acidic to the most basic.
- 3 Record which substances, if any, were neutral.

Discussion

- 1 Identify any significant differences between the pH values obtained using universal indicator and the pH meter. Suggest possible reasons for these differences.
- 2 Suggest a procedure for measuring the pH of the soil samples using a pH meter.
- 3 Identify any noticeable difference in the pH of the water samples and suggest a reason for this difference.
- 4 Compare the hydrogen ion concentration of the known samples of HCl used with the pH reading obtained from the meter. Suggest what a difference in these readings could imply.
- 5 Reflect on possible errors which could have occurred and suggest what effect these may have had on the results.

Conclusion

Identify any generalisations related to the acidity and alkalinity of the substances tested.



REVISION

Take this quiz to check your understanding of acids, bases and pH.

Calculations involving acids and bases

You have already learnt about the relationship between concentration, moles and volume of a solution. You also learnt about dilutions. We will now apply this information to acids, bases and pH.

WORKED EXAMPLE 7.5

- a How many moles of calcium oxide would be needed to neutralise 100 mL of a 0.400 mol L^{-1} solution of HCl?
- b What mass of calcium oxide would need to be added to the HCl solution?
- c What is the pH of the final solution?
- d If a solution of HNO_3 with a pH of 3 was diluted by a factor of 100, what is the pH of the final solution?

Answers

- a 0.0200 mol are needed.

Logic

- 1 Write a balanced equation.



- 2 Calculate the number of moles of HCl using $C = \frac{n}{V}$.

$$c = 0.400 \text{ mol L}^{-1}, n = ?, V = 100 \text{ mL} = 0.100 \text{ L}$$

$$0.400 = \frac{n}{0.100}$$

$$n(\text{HCl}) = 0.400 \times 0.100 = 0.0400 \text{ mol}$$



- 3 Use the balanced equation to calculate the number of moles of CaO needed to react with the HCl.

$$n(\text{CaO}) = \frac{1}{2} n(\text{HCl})$$

$$n(\text{CaO}) = \frac{0.0400}{2} = 0.0200 \text{ mol}$$

- b The mass is 1.12 g

- 4 Calculate the mass of CaO.

$$n = \frac{m}{M}$$

$$n = 0.0200 \text{ mol}, m = ?, M(\text{CaO}) = 56.08 \text{ g mol}^{-1}$$

$$0.0200 = \frac{m}{56.08}$$

$$m = 0.0200 \times 56.08 = 1.12 \text{ g}$$

- c The pH is 7.

- 5 Determine the pH.

Since the reaction involved using an exact quantity of CaO to neutralise the HCl, the final solution should be neutral. Therefore, the pH should be 7.

- d The final pH is 5.

Recall the relationship between pH and concentration. A change by a factor of 10 in concentration is equivalent to a change in 1 unit in the pH scale.

Therefore, a change in concentration by a factor of 100 means a change in pH by two units.

The solution is being diluted so the pH will increase and the pH of the final solution will be 5.

Try these yourself

- How many grams of KOH are needed to neutralise 500 mL of $2 \text{ mol L}^{-1} \text{ H}_2\text{SO}_4$?
- How many grams of $\text{Mg}(\text{OH})_2$ are needed to neutralise 1.00 L of $0.25 \text{ mol L}^{-1} \text{ HNO}_3$?
- If a solution of HCl with a pH of 1 is diluted by a factor of 1000, what will the pH of the diluted solution be?
- A 500 mL solution of NaOH of pH 9 was left to evaporate until the resulting volume was 50 mL. What is the pH of the resultant solution?

QUESTION SET 7.4

Remembering

- Define 'pH'.
 - Explain what information the pH scale provides.
- What is an indicator?
 - What is the purpose of an indicator?
- True or false? If a statement is false, rewrite it so it is correct.
 - An acid tastes sour while a base tastes bitter.
 - A soluble base is called an acid.
 - A substance that is neither acid nor base is negative.
 - Bases have the power to neutralise acids.
 - An acid turns litmus blue while a base turns litmus red.

Understanding

- What is the relationship between concentration of an acid, concentration of a base and pH?
- An ant bite contains an acid while a wasp sting contains a base. Choose which of vinegar or ammonia would be best to treat each type of sting and explain why.

- 6 Consider the pH ranges 1–3, 4–6, 7, 8–10 and 11–13.
Which pH range would apply to each of the following solutions?
- a Concentrated hydrochloric acid
 - b Distilled water
 - c Dilute ammonia solution
 - d Oven cleaner
- 7 If a 100 mL container of orange juice concentrate had a pH of 5 and the instructions on the container were to add the concentrate to 900 mL of water, what would be the pH of the diluted solution?

Applying

Use Figure C7.16 to answer the following question.

- 8 Which of the following statements is definitely correct?
- A A substance that turns phenolphthalein pink is more acidic than one that turns methyl orange pink.
 - B A substance that turns methyl orange yellow is less acidic than one that turns methyl orange pink.
 - C A substance that turns phenolphthalein colourless is neutral.
 - D A substance that turns litmus red is more acidic than one that turns bromothymol blue a blue colour.
- 9 Which of the statements in Question 8 could possibly be correct, depending on the pH of the solution. Give the pH range that would make these statements correct.

Analysing

- 10 An aquarium was at the correct pH when a young child poured a 500 mL bottle of vinegar into it. The label on the bottle showed the vinegar in the bottle to be a 0.83 mol L^{-1} solution of acetic (CH_3COOH) acid.
- a How had the pH of the aquarium changed as a result of this action?
 - b How many grams of sodium hydroxide would need to be added to the aquarium to neutralise excess acid?



Science Photo Library/Martyn F. Chillmaid

Figure C7.18 ▲

The reaction between magnesium metal and dilute hydrochloric acid

7.6 Reactions of acids

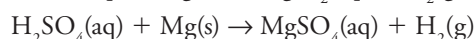
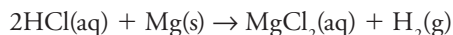
Acids undergo a number of different types of reactions. As with acid–base reactions, once the pattern of the reaction is known, products for most reactions of these types can be predicted.

Reactions between acid and a metal

Acids react with metals. When rain becomes acidic due to pollutants in the air, it reacts with metals, causing them to corrode more quickly. Not all metals are reactive, but for those that are the reaction can be represented by the general equation:



The following reactions are examples of reactions between an acid and magnesium metal.



This reaction could also be called a corrosion reaction because the metal is eaten away.

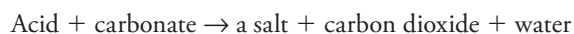


Reaction between an acid and a carbonate or hydrogen carbonate

Many products such as bread and sponge cakes have a honeycomb structure, which contains bubbles. These bubbles are formed by carbon dioxide gas causing the mixture to rise during cooking. Carbon dioxide is formed by adding a carbonate (CO_3^{2-}) or hydrogen carbonate (HCO_3^-) and an acid to the cooking mixture.

When an acid and carbonate react, they produce a salt, carbon dioxide and water.

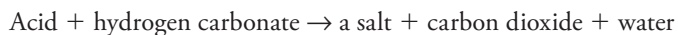
This can be represented by the general equation:



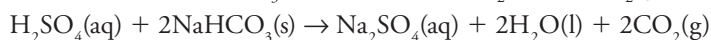
The following reactions are examples of reactions between an acid and calcium carbonate.



When an acid and a hydrogen carbonate react, they will also produce a salt, water and carbon dioxide according to general equation:



The following reactions are examples of reactions between an acid and sodium hydrogen carbonate.



There is a pattern to all these reactions. Knowing the pattern means the products can be predicted.



Mark Fergus Photography

Figure C7.19 ◀
Marble chips (calcium carbonate) react vigorously with hydrochloric acid.

QUESTION SET 7.5

Remembering

1 Complete the following general equations for the reactions of acids.

- a Acid + base $\rightarrow$
- b Acid + metal $\rightarrow$
- c Acid + carbonate $\rightarrow$

Understanding

2 Using the general reactions as a guide, write balanced equations for the following reactants.

- a $\text{H}_2\text{SO}_4(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow$
- b $\text{CH}_3\text{COOH}(\text{aq}) + \text{Ca}(\text{OH})_2(\text{aq}) \rightarrow$
- c $\text{HNO}_3(\text{aq}) + \text{Mg}(\text{s}) \rightarrow$
- d $\text{HCl}(\text{aq}) + \text{Zn}(\text{s}) \rightarrow$
- e $\text{HF}(\text{aq}) + \text{Ca}(\text{HCO}_3)_2(\text{s}) \rightarrow$
- f $\text{HNO}_3(\text{aq}) + \text{CuCO}_3(\text{s}) \rightarrow$

7.7 Identifying unknown ions in solution

In Experiment 7.1, you used your understanding to identify ions in unknown solutions. There are several common tests that can be used to identify particular ions in solution. The experimental results are then compared with known test results and this allows ions to be identified.

Table C7.5 Tests used to identify the four selected anions

Anion	Test
Carbonate (CO_3^{2-})	1 Solution has a pH between 8 and 11 (pH paper suffices). 2 Addition of dilute HNO_3 produces bubbles of colourless gas (CO_2).^A
Sulfate (SO_4^{2-})	1 Addition of $\text{Ba}(\text{NO}_3)_2$ to an acidified sample of the solution produces a thick white precipitate. 2 Acidification and addition of $\text{Pb}(\text{NO}_3)_2$ produces a white precipitate.
Phosphate (PO_4^{3-})	1 Addition of ammonia followed by $\text{Ba}(\text{NO}_3)_2$ produces a white precipitate. 2 Addition of Mg^{2+} in an ammonia/ammonium nitrate buffer produces a white precipitate, $\text{Mg}(\text{NH}_4)\text{PO}_4$. 3 Acidification with HNO_3 followed by addition of ammonium molybdate solution ($(\text{NH}_4)_2\text{MoO}_4$) produces a yellow precipitate; warming the mixture for a few minutes may be necessary.
Chloride (Cl^-)	1 Addition of AgNO_3 to an acidified sample produces a white precipitate,^B which dissolves in ammonia solution and darkens in sunlight.

^A Any strong acid would do, but for analysing mixtures we do not want to introduce any Cl^- or SO_4^{2-} .

^B In non-acidic solutions, silver nitrate also produces precipitates with carbonate and phosphate (and with sulfate at all pH values if sulfate concentration is moderately high), so this test alone does not prove the presence of chloride: it is also necessary to prove the absence of sulfate.

Table C7.6 Precipitates formed between cations and anions

Cation ^A	Anion					
	OH ⁻	Cl ⁻	SO ₄ ²⁻	CO ₃ ²⁻ ^B in alkaline solution	PO ₄ ³⁻	
					Solution pH < 2	Solution pH > 6
Ba ²⁺	No	No	Yes	Yes	No	Yes
Pb ²⁺	Yes	Yes ^C	Yes	Yes	No	Yes
Ag ⁺	Yes ^D	Yes	Yes ^C	Yes	No	Yes
Cu ²⁺	Yes	No	No	Yes	No	Yes

^A In acidic or alkaline solution.^B Cannot have carbonate in acid solution: it decomposes to CO₂(g).^C Provided concentration of Cl⁻ or SO₄²⁻ is not too low, say > 0.05 mol L⁻¹.^D Precipitate is brown Ag₂O.

In Experiment 7.4, you will use the information provided in Tables C7.5 and C7.6, along with what you have learnt, to identify some unknown anions. A similar process could be used to identify cations.

In addition, cations can be identified by flame tests, which you conducted in Experiment 1.1 on page 123.

Table C7.7 Tests used to identify cations

Cation	Tests
Ag ⁺	With Cl ⁻ forms a white precipitate that dissolves in ammonia solution
Pb ²⁺	With Cl ⁻ forms a white precipitate (if solution not too dilute, say >0.05 mol L ⁻¹ and this precipitate does not dissolve in ammonia solution) With I ⁻ forms a yellow precipitate
Ba ²⁺	With SO ₄ ²⁻ forms a white precipitate Gives a pale green flame colour No precipitate with OH ⁻ or F ⁻ (compare Ca ²⁺)
Ca ²⁺	With SO ₄ ²⁻ forms a white precipitate (if solution not too dilute, say >0.05 mol L ⁻¹) With F ⁻ forms a white precipitate Gives a brick-red flame colour
Zn ²⁺	With hydroxide forms a white precipitate This precipitate dissolves in both excess OH ⁻ solution and ammonia solution
Cu ²⁺	With OH ⁻ forms a blue precipitate This precipitate dissolves in NH ₃ to form a deep blue solution Gives a blue-green flame colour
Fe ²⁺	With OH ⁻ forms a green or white precipitate, which may turn brown Decolorises acidified dilute potassium permanganate solution
Fe ³⁺	With OH ⁻ forms a brown precipitate With thiocyanate (SCN ⁻) forms a deep red solution

EXPERIMENT 7.4

IDENTIFYING UNKNOWN ANIONS

In this experiment you will conduct a series of tests to identify specified anions in a solution that contains only one anion. Many of the reactions that occur in this experiment are quite complicated, so you are not expected to write equations for them. The main purpose of this experiment is to understand the processes used.

Aim

To carry out a series of chemical reactions to devise chemical tests for identifying the phosphate, sulfate, carbonate and chloride anions (PO_4^{3-} , SO_4^{2-} , CO_3^{2-} and Cl^-) in solution when these are the only ions that could be present

Materials

- Dropper bottles containing 0.1 mol L^{-1} of test solutions $\text{Pb}(\text{NO}_3)_2$, $\text{Ba}(\text{NO}_3)_2$, HNO_3 , AgNO_3 and NaOH
- Dropper bottles containing 0.1 mol L^{-1} of anion solutions Na_2SO_4 , Na_2CO_3 , Na_3PO_4 and NaCl
- 4 test tubes and test-tube rack
- Distilled water
- Water bath

What are the risks in doing this experiment?	How can you manage risks to stay safe?
Concentrated NaOH and HNO_3 are corrosive.	Wear safety glasses and protective clothing. Do not allow contact with skin or clothes. If contact occurs, wash with large amounts of water for 10–15 minutes.
Some metal salts (silver, barium and lead) are poisonous. Silver nitrate will leave a brown stain on clothing and skin.	Avoid contact with skin and clothes. Wash hands thoroughly after use. Dispose of the chemicals as directed by your teacher.

In your write-up, add any more risks you can think of, as well as ways to manage them. In particular, expand on the two risks listed to identify specific risks involved with each of them.

Procedure

- 1 Add 10 drops of each of the anion solutions to each of four test tubes.
- 2 Add 10 drops of HNO_3 to each of the test tubes and warm them gently.
- 3 Record the results.
- 4 Use these results to describe a test for one particular anion.
- 5 Thoroughly clean all test tubes with distilled water between tests.
- 6 For the remaining three anions, put 10 drops in each of three test tubes, add 5 drops of HNO_3 and then add 5 drops of Ag^+ .
- 7 Record the results in a table like the one in the results section. Write 'NP' if no precipitate forms and 'ppt' if precipitate forms and give its colour.
- 8 Repeat steps 6 and 7 with Pb^{2+} and then Ba^{2+} .
- 9 After testing the Ba^{2+} , add 10 drops of NaOH to each of the test tubes and record any changes.

Extension

- 1 Use the results from the tests to identify the anion (one per solution) in each of three unknown solutions.
- 2 Try to identify an unknown solution containing two anions.

Results

Test solution/anion	H ⁺	Ag ⁺	Pb ²⁺	Ba ²⁺	Ba ²⁺ and OH ⁻
CO ₃ ²⁻					
Cl ⁻					
SO ₄ ²⁻					
PO ₄ ³⁻					

Analysis of results

For each test and result below, decide which anion would be the most appropriate fit to these results.

Anion	Test and results
	Gives a ppt with acidified Ag ⁺ but not with Ba ²⁺
	Gives a ppt with acidified Ba ²⁺
	Gives a ppt with Ba ²⁺ in alkaline solution but not acid solution
	Produces bubbles of gas with addition of dilute HNO ₃

Discussion

- 1 Is the lead ion useful in identifying anions? Explain why or why not.
- 2 When identifying anions in a solution in which two anions may be present, it is necessary to destroy any CO₃²⁻ before further testing for other anions can be carried out. How might this be done? (Consider the tests in this experiment.) Why is this necessary?

Conclusion

How effective was your identification?



WATER QUALITY

Visit this virtual lab in which you will use analysis to investigate water quality.

CHAPTER CHECKLIST

You should know:

- the different types of equations that can be used to show precipitation reactions
- concentration is a measure of the amount of solute in a given volume of solution
- different units of concentration
- how dilution affects concentration
- the relationship between the hydrogen and hydronium ion
- the general equation for the neutralisation reaction between an acid and base
- the purpose of an indicator
- the relationship between pH, hydrogen ion and hydroxide ion concentrations
- the general reactions for acid and metal, acid and carbonate, acid and hydrogen carbonate
- the definition of precipitate, acid and base
- the definition of pH and pH scale.

You should be able to:

- define precipitate, acid and base
- define pH and the pH scale
- use solubility tables to predict the formation of a precipitate
- write balanced reactions showing precipitation
- use experimental data to identify unknown ions
- calculate concentration
- convert between moles, volume, concentration and mass
- calculate mass of reactants and products in a chemical reaction given the amount of one substance in the reaction
- calculate concentrations and volumes for dilutions
- identify an acid and a base
- predict the products of acid–base reactions
- measure pH and use data to determine pH
- predict products of reactions
- use analytical techniques to identify specific ions.

CHAPTER GLOSSARY

acid a substance that produces hydrogen ions (H^+) in aqueous solution

alkali a soluble base

amphoteric a substance that can act as either an acid or a base depending on the other reactants

aqueous solutions solutions in which water is the solvent

base a substance that either contains the oxide (O^{2-}) or hydroxide ion (OH^-) or produces the hydroxide ion in aqueous solution

complete ionic equations equations that show all the ions present in solution

concentration the amount of solute dissolved in a given volume of solvent

dilution the process of adding water to a solution to reduce its concentration

hydronium ion (H_3O^+) the ion produced when a hydrogen ion (H^+) attaches to a water molecule

indicator a substance that changes colour in solution depending on whether the solution is acidic or basic

molarity the number of moles of solute per litre of solution

net ionic equations an equation that only contains the ions that take part in the chemical reaction; that is, spectator ions are excluded

neutral neither acidic nor basic, pH of 7

neutralisation a reaction between an acid and a base to form a salt and water

pH scale a scale that measures the acidity and alkalinity of a solution, based on the concentration of hydrogen ions in a solution

precipitate a solid produced by reaction between two clear solutions

precipitation reaction a reaction in which a precipitate is produced

solubility the degree to which a substance dissolves

spectator ion an ion that is present in the same form on each side of the equation

universal indicator a mixture of different indicators that displays a range of colours when added to solutions of different pH

volumetric flask a piece of glass laboratory equipment used to prepare solutions of precise concentration

CHAPTER REVIEW QUESTIONS

Remembering

- 1 Write the definition of:
 - a precipitate.
 - b acid.
 - c base.
 - d pH.
 - e concentration.
- 2 What is a spectator ion?
- 3 What is the relationship between the hydrogen ion and the hydronium ions?
- 4 What is the relationship between pH, hydrogen ion concentration and hydroxide ion concentration?

Understanding

- 5 Use the solubility table to determine whether or not a precipitation reaction will occur when the following reactants are mixed. Where a reaction occurs, write the balanced overall equation and net ionic equation for the reaction.
 - a Potassium chloride and zinc nitrate
 - b Ammonium bromide and lead nitrate
 - c Ammonium sulfide and magnesium acetate
 - d Strontium chloride and zinc sulfate
- 6 What solutions would you mix to produce each of the following precipitates?
 - a Lead sulfate
 - b Iron(II) sulfide
 - c Magnesium hydroxide
- 7 How many moles of solute do you need to make the following solutions?
 - a 2.00 L of 1.50 mol L^{-1} sodium chloride
 - b 3.5 L of 0.20 mol L^{-1} potassium hydroxide
- 8 Calculate the molarity of the solutions made by dissolving the following amounts of solute in water and making the volume up to the stated value.
 - a 5.0 mol nitric acid in 2.0 L
 - b 2.5 mol sodium hydroxide in 0.50 L
- 9 What mass of the indicated substance do you need to weigh out to make the following solutions?
 - a Barium hydroxide to make 0.5 L of 0.060 mol L^{-1} hydroxide ion solution
 - b Sulfuric acid to make 250 mL of 0.330 mol L^{-1} hydrogen ion solution

- 10 The volume of solution in column A was diluted to the volume in column B. Calculate the molarity of the diluted solution.

	Column A	Column B
a	50 mL of 0.242 mol L ⁻¹ hydrochloric acid	500 mL
b	25 mL of 0.152 mol L ⁻¹ sulfuric acid	2.0 L
c	10 mL of 0.114 mol L ⁻¹ sodium hydroxide	250 mL

- 11 What volume of solution in Column A in the following table is needed to prepare the solution in Column B?

	Column A	Column B
a	0.282 mol L ⁻¹ hydrochloric acid	250 mL of 0.0113 mol L ⁻¹
b	2.42 mol L ⁻¹ sulfuric acid	2.0 L of 0.121 mol L ⁻¹
c	0.318 mol L ⁻¹ sodium hydroxide	1.0 L of 0.300 mol L ⁻¹

- 12 What colour would you expect to see in a piece of litmus paper to which the following have been added?

- a 1 drop of Ca(OH)₂(aq)
- b 1 drop of HF(aq)
- c 1 drop of NaNO₃(aq)

- 13 Milk has a pH of 6.5. This means that milk is:

- A strongly acidic.
- B slightly acidic.
- C slightly basic.
- D strongly basic.

- 14 If tomato juice has a pH of 5 and orange juice has a pH of 3, then:

- A orange juice is twice as acidic as tomato juice.
- B tomato juice is twice as acidic as orange juice.
- C orange juice is 100 times more acidic than tomato juice.
- D tomato juice is 100 times more acidic than orange juice.

- 15 Using the general reactions as a guide, write balanced equations for the following.

- a HNO₃(aq) + KOH(aq) →
- b HCl(aq) + Al(OH)₃(aq) →
- c H₂SO₄(aq) + Fe(s) →
- d CH₃COOH(aq) + MgCO₃(s) →
- e H₃PO₄(aq) + NaHCO₃(s) →

Applying

- 16 A solution was thought to contain either lead nitrate or calcium nitrate. Describe what tests could be carried out to correctly identify the substance.
- 17 A solution was known to contain either sodium carbonate or sodium chloride. Samples of the solution gave bubbles of gas with HNO₃ but no precipitate with AgNO₃. Which anion is present? Explain.
- 18 A solution was thought to contain either barium chloride or calcium chloride. Samples of the solution gave a white precipitate with Na₂SO₄ and a pale green flame colour. Which of the ions is present? Explain.
- 19 If a 1 L container of HCl had a pH of 3 and instructions on the container were to dilute 100 mL to 1 L before use, then what would be the pH of the solution being used?

- 20** To determine the solubility of calcium hydroxide ($\text{Ca}(\text{OH})_2$) a chemist took 25 mL of a saturated calcium hydroxide solution and found that it reacted completely with 8.13 mL of 0.102 mol L^{-1} hydrochloric acid. Calculate the:
- molarity of the calcium hydroxide solution.
 - solubility of calcium hydroxide in grams per litre.
- 21** In 16.4 mL of 0.117 mol L^{-1} magnesium chloride solution, how many moles are there of:
- magnesium chloride?
 - magnesium ions?
 - chloride ions?

Analysing

- 22** A student was given four solutions and told each contained one of the following cations: Pb^{2+} , Cu^{2+} , Ba^{2+} , Ca^{2+} , Fe^{2+} or Fe^{3+} . The results of a series of tests that were conducted are given in the following table, where NP stands for no precipitate and ppt stands for precipitate.

Reagent added	A	B	C	D
KI	Yellow ppt	NP	NP	NP
H_2SO_4	ppt	NP	ppt	ppt
NaOH	ppt	Brown ppt	NP	ppt

Use the given results to identify possible cations for A–D. If there is not enough information, suggest other possible tests and their expected results that could be used to identify any still unknown solutions.

- 23** A solution was made by dissolving 22.22 g of calcium chloride in enough water to make a 500 mL solution. This was then reacted with 0.5 mol L^{-1} sulfuric acid to produce insoluble calcium sulfate. Calculate the:
- volume of sulfuric acid needed to react
 - mass of calcium sulfate produced
- 24** Consider the following table of different indicators showing colour change over different pH ranges.

Indicator	Colour change				
	Highly acidic	Slightly acidic	Neutral	Slightly alkaline	Highly alkaline
Methyl orange	Red	Yellow	Yellow	Yellow	Yellow
Bromothymol blue	Yellow	Yellow		Blue	Blue
Litmus	Red	Red	Purple	Blue	Blue
Phenolphthalein	Colourless	Colourless	Colourless	Colourless	Red

- Solution A is red in methyl orange, while solution B is red in phenolphthalein. Which is the more acidic?
- Four different solutions were tested with different indicators. Which of the solutions could be neutral?
 - Colourless in phenolphthalein
 - Red in litmus
 - Yellow in methyl orange
 - Blue in bromothymol blue

25 The following table shows the most suitable pH ranges for the growth of some common plants.

Flower	pH range	Crop	pH range
Azalea	4.5–5.5	Barley	6.0–8.0
Calendula	6.0–7.5	Clover	5.5–7.0
Hibiscus	6.0–7.0	Wheat	5.5–7.5
Daffodil	6.0–6.5	Cotton	5.0–6.0
Sweet pea	7.0–8.0		

The pH of a number of soils was tested and the results are shown below.

Soil	pH
A	4.0
B	5.0
C	6.0
D	7.5
E	8.5

- a** Which soil was most basic?
- b** In which soil(s) would clover probably grow best?
- c** Which crop is most likely to grow best in soil A? Why?
- d** What would need to be added to soil B to make it suitable for growing barley?
- e** Hydrangea flowers change colour depending on the soil pH. They have blue flowers in acidic soil and pink flowers in basic soil. If some were planted in soil B, then what coloured flowers would they produce?

Reflecting

26 A chain of coffee shops started giving away brightly coloured drinking mugs. It was thought that lead compounds had been used in the paint and that this dissolved in acidic drinks such as soft drinks and fruit juices.

Design an experiment to determine:

- a** whether lead is present in the paint.
- b** the quantity of lead dissolved in 100 mL of an acidic drink.